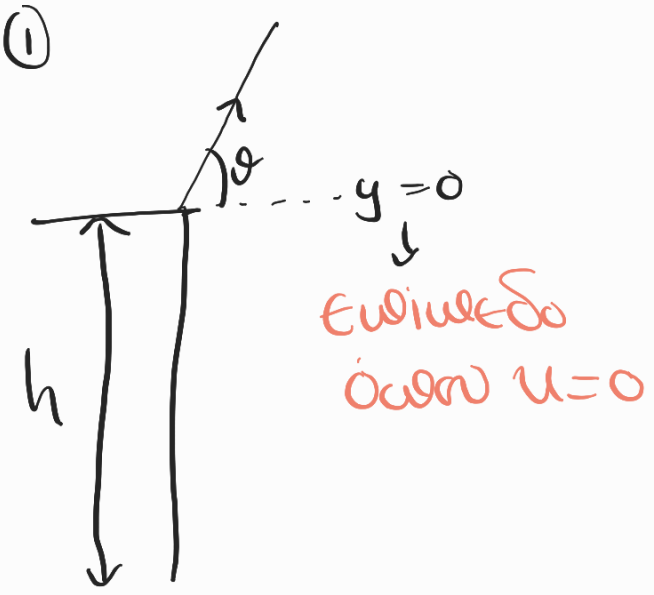
①

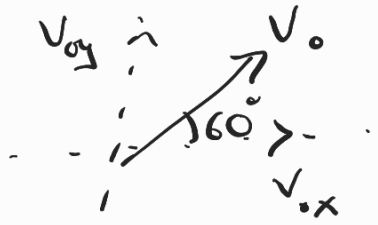


ευθύγραμμο ομαλά $u=0$

$V_0 = 16 m/s$

$h = 12m$

$\vartheta = 60°$



$$E_0 = \frac{1}{2} m V_0^2$$

$$E_{top} = \frac{1}{2} m v^2 + mgy$$

$$\Rightarrow E_0 = E_{top} \Rightarrow \frac{1}{2} m V_0^2 = \frac{1}{2} m v^2 + mgy$$

$$\Rightarrow gy = \frac{V_0^2}{2} - \frac{v^2}{2}$$

$$\Rightarrow y = \frac{1}{2g}(V_0^2 - v^2)$$

$$v = V_{0x} = V_0 \cos\vartheta \Rightarrow y = \frac{V_0^2 - V_0^2 \cos^2\vartheta}{2g} = 9.8m$$

β) $$E_f = \frac{1}{2} m V_f^2 + mgy$$

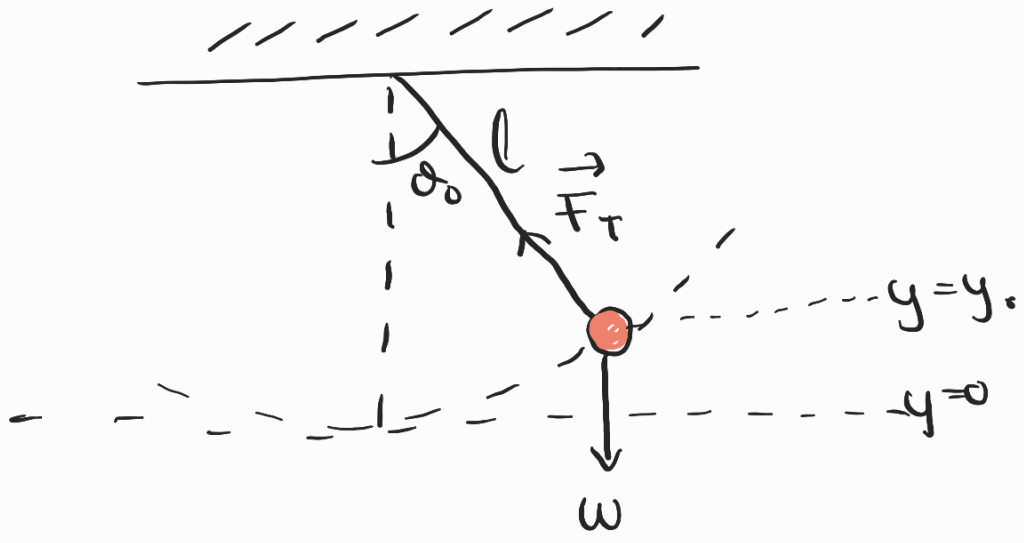
$$E_f = E_0 \Rightarrow \frac{1}{2} m V_0^2 = \frac{1}{2} m V_f^2 + mgy$$

$$\Rightarrow \frac{V_f^2}{2} = \frac{V_0^2}{2} - gy$$

$$\Rightarrow V_f^2 = V_0^2 - 2gy \Rightarrow v_f = \sqrt{V_0^2 - 2gy}$$

$y_f = -12m \Rightarrow v_f = \sqrt{16^2 - 2(10)(-12)} = 22 m/s$ //

(2)



Α. Για $t=0, y=y_0 \Rightarrow$ η ταχύτητα της σφαίρας είναι $v=0$ άρα $E_K=0$
Η U είναι μέγιστη αφού $y=y_{max}$

Για $\vartheta_0 > \vartheta > 0$:

Η E_K αυξάνεται και γίνεται μέγιστη στο $y=0$ όταν $v=0$, ενώ $U=0$ εφόσον είμαστε στο επίπεδο μηδενικής δυναμικής ενέργειας.

Για $0 < \vartheta < -\vartheta_0$: Η E_K μειώνεται και αυξάνεται η U. Στο $\vartheta = -\vartheta_0$

η E_K είναι 0 εφόσον $v=0$ και η U είναι μέγιστη.

B. θέση ϑ:

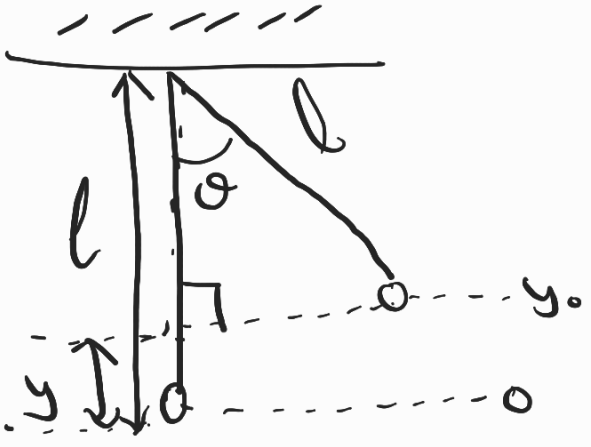
$$\Rightarrow E_\vartheta = \frac{1}{2}mv^2 + mgy$$

ϑέση $\vartheta = \vartheta_0$:

$$\Rightarrow E_{\vartheta_0} = mgy_0$$

ΑΔΕ: $E_\vartheta = E_{\vartheta_0} \Rightarrow \frac{1}{2}mv^2 + mgy = mgy_0$

$$\Rightarrow \frac{1}{2}v^2 + gy = gy_0 \quad (1)$$



$$l - y = l\cos\vartheta$$

$$\Rightarrow y = l - l\cos\vartheta \quad (2)$$

$$\Rightarrow y_0 = l - l\cos\vartheta_0 \quad (3)$$

(2), (3) → (1): $\frac{v^2}{2} + g(l - l\cos\vartheta) = g(l - l\cos\vartheta_0)$

$\Rightarrow \frac{v^2}{2} + gl - gl\cos\vartheta = gl - gl\cos\vartheta_0$

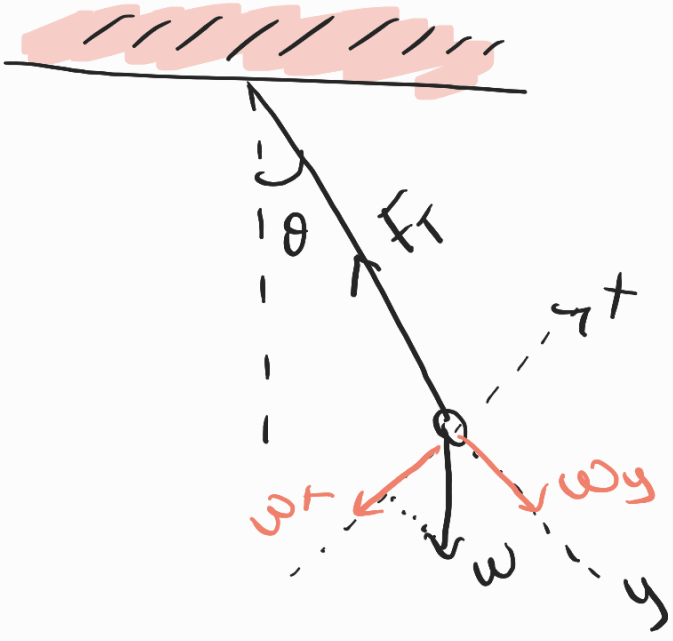
$\Rightarrow v^2 = 2gl(\cos\vartheta - \cos\vartheta_0)$

$\Rightarrow v = \sqrt{2gl(\cos\vartheta - \cos\vartheta_0)}$ //

Γ. $y = 0 \Rightarrow \vartheta = 0$

$\Rightarrow v = \sqrt{2gl(1 - \cos\vartheta_0)}$

Δ.



$\Rightarrow W_x = mg\sin\vartheta$

$W_y = mg\cos\vartheta$

$$\Sigma F_y = m a_{rad} \Rightarrow F_T - W_y = \frac{m\upsilon^2}{r}$$

$$\Rightarrow F_T - mg\cos\vartheta = \frac{m\upsilon^2}{r} \quad \text{αλλά } r = \ell$$

$$\Rightarrow F_T = mg\cos\vartheta + m\upsilon^2/\ell \quad (4)$$

αφού $B: \upsilon = \sqrt{2g\ell(\cos\vartheta - \cos\vartheta_0)}$ (5)

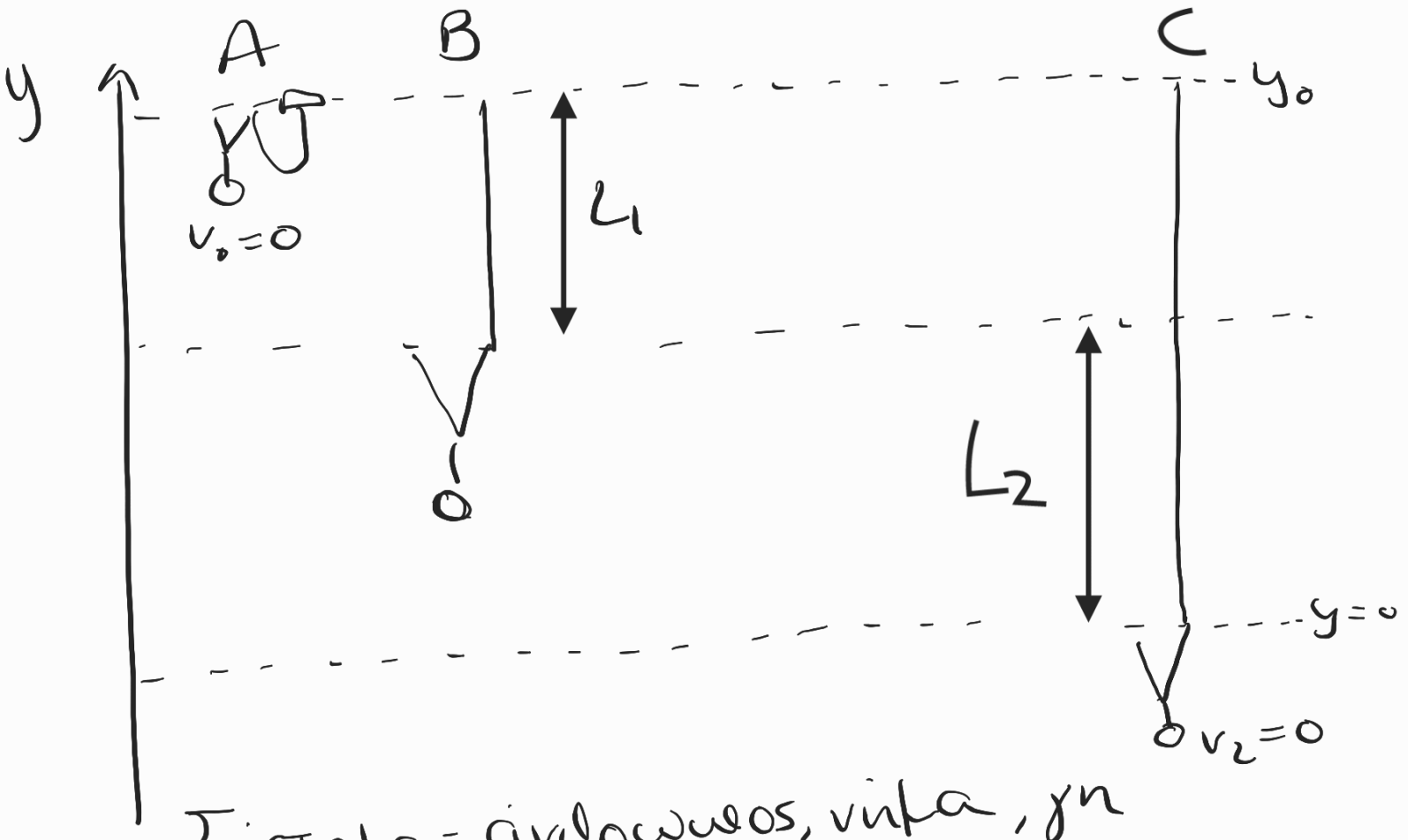
$(5) \rightarrow (4): F_T = mg\cos\vartheta + \frac{2mg\ell(\cos\vartheta - \cos\vartheta_0)}{\ell}$

$$\Rightarrow F_T = 3mg\cos\vartheta - 2mg\cos\vartheta_0$$

$$\Rightarrow F_T = mg(3\cos\vartheta - 2\cos\vartheta_0)$$

Για $\vartheta = \vartheta_0 \Rightarrow \boxed{F_T = mg\cos\vartheta_0}$ //

(3) $a_0 = ?$



Σώστημα = άνθρωπος, γη

$y_0 = 134\,m$

$m = 100\,k$

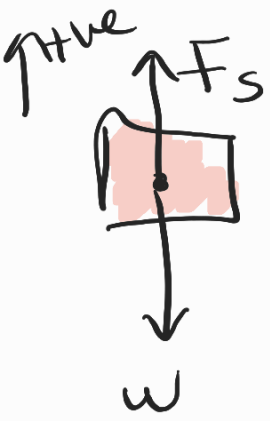
$E_C = U_s + U_g + K = \frac{1}{2} k L_2^2$

$E_A = U_s + U_g + K = mgy_0$

$E_C = E_A \Rightarrow \frac{1}{2} k L_2^2 = mgy_0$

$\Rightarrow k = \frac{2mgy_0}{L_2^2} = \frac{2(100)(10)(134)}{(80)^2}$

$\Rightarrow k = 41.9\ N/m$ //



$F_s \equiv$ δύναμη ελατηρίου

→ διάγραμμα δυνάμεων στο $y = 0\ m$

$$\Rightarrow F_s - W = m \cdot a$$

$$\Rightarrow k l_2 - mg = m \cdot a$$

$$\Rightarrow a = \frac{k l_2}{m} - g$$

$$\Rightarrow a = \frac{(41.9)(80)}{100} - 10 =$$

$$\Rightarrow a = 33.52 - 10 = 23.52\ m/s^2$$

④ (α) A : − D : 0

B : 0 E : −

C : + F : 0

$$F = -\frac{dU}{dx} = -\text{κλίση γραφήματος}$$

(β) $|F_{max}| \Rightarrow$ μέγιστη κλίση

$\Rightarrow$ C

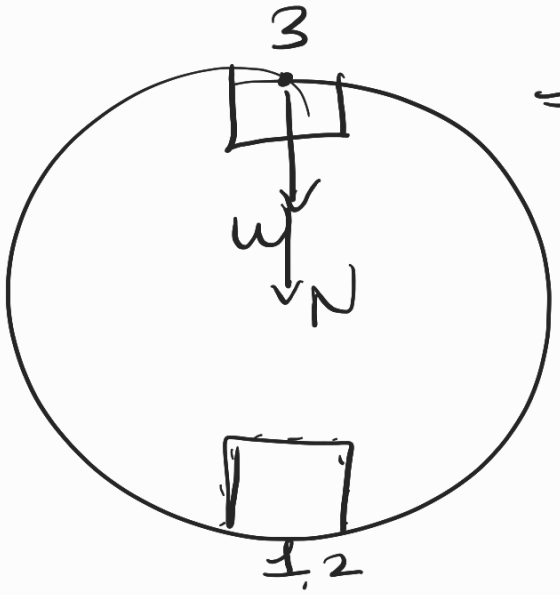
(γ) B : ασταθής

D : σταθερή

F : ουδέτερη

⑤ R αυτινα βρόχων

$\Rightarrow y_0 = 4 \cdot R$



$\Rightarrow N + w = \frac{m v_3}{R}$ ①

$E_0 = mg \cdot 4R$
$E_1 = \frac{1}{2} m v_1^2$
$\Rightarrow 4Rmg = \frac{1}{2} m v_1^2$
$\Rightarrow v_1 = \sqrt{8Rg}$

Μετά τον βαλιο : $v_2 = 0.75 v_1$

$\Rightarrow v_2 = 0.75\sqrt{8Rg}$

$E_3 = \frac{1}{2} m v_3^2 + mg2R$

$E_2 = \frac{1}{2} m v_2^2$

$\Rightarrow E_3 = E_2$

$$\Rightarrow \frac{1}{2}mv_3^2 + mg2R = \frac{1}{2}m\left(0.75\sqrt{8Rg}\right)^2$$

$$\Rightarrow \frac{v_3^2}{2} = \frac{0.75^2}{2}\cdot 8Rg - 2Rg$$

$$\Rightarrow v_3^2 = 0.75^2 \cdot 8Rg - 4Rg$$

$$\Rightarrow v_3^2 = (0.75^2 \cdot 8 - 4)Rg$$

$$\Rightarrow v_3 = 0.5Rg \quad (2)$$

$$(2) \rightarrow (1): N + mg = \frac{m(0.5Rg)}{R}$$

$$\Rightarrow N = -0.5mg$$

$\Rightarrow$ υάθετη αντίδραση

αρνητική άρα το τρενάκι

φεύγει από τις ράγες!